

- 24** The diagram shows a car travelling at a constant speed in a straight line between person P and person Q from point X to point Y.



The car sounds its horn continuously as it travels. The horn emits sound of constant frequency.

Which statements about what person P and person Q hear during the motion of the car are correct?

- 1 Person P hears a sound of increasing frequency.
- 2 Person Q hears a sound of decreasing frequency.
- 3 Person Q always hears a sound of higher frequency than person P.

- A** 1, 2 and 3
B 1 and 2 only
C 3 only
D none of them

- 25** A progressive wave of frequency 200 Hz is travelling with a speed of 600 m s^{-1}